

Bill of Materials

Component	Value/Part	Qty	Approx Price	Source
ESP32-C3	ESP32-C3	1	₹261	Robu.in
MPU6050	MPU6050 module	1	₹169	Robu.in
HC-SR04	Ultrasonic sensor	2	₹69 each	Robu.in
Resistor	4.7k Ω	2	₹1	Local Store
Resistor	330 Ω	1	₹1	Local Store
Capacitor	0.1 μ F	2	₹2	Local Store
LED	Red LED	1	₹5	Local Store
LiPo Battery	3.7V	1	₹198	Robu.in
Jumper Wires	Wires	20	₹36	Local Store

Estimated Total Cost= ₹811

5-line answer: Your walking stick used ATtiny85. For a wearable with WiFi alerts, would you keep ATtiny85 or switch to ESP32? Why?

- I would switch to ESP32 for a wearable. ESP32 has built in WiFi and Bluetooth which is useful for wireless alerts. It also has faster processing capability and more number of GPIO pins enables us to have multiple sensors. It also supports I2C communication easily and operates at 3.3V making it suitable for a wearable device. Though ATtiny85 consumes lesser power, ESP32 is more ideal for a wearable device.